

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (EE) Theory Examination****March – April 2018****EE304.01/EE304 Industrial Instrumentation****Date: 29.03.2018, Thursday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q – 1.a** Discuss the various classifications of transducers. **[05]****Q – 1.b** Define the followings: **[04]**

- | | |
|--------------|------------------|
| 1. Precision | 2. Repeatability |
| 3. Accuracy | 4. Calibration |

Q – 2 Attempt any two questions. **[12]**

- (i) Draw a schematic of total radiation pyrometer.
- (ii) With neat sketch, explain the operating principle and construction of thermocouple.
- (iii) Draw a schematic of resistive method for measurement of liquid level.

Q - 3 Attempt any two questions. **[14]**

- a. Explain dew point meter in detail.
- b. Explain the rotary voltage differential transformer (R.V.D.T.) along with its advantages and disadvantages.
- c. Explain infra-red pyrometers.

SECTION – II**Q - 4** Draw a sketch of flow nozzle used to measure the flow rate of the fluid. **[05]****Q – 5.a** Explain the pitot tube used for flow measurement along with its advantages and disadvantages. **[06]****OR****Q – 5.a** Explain the variable inductance type strain gauge transducer. **[06]****Q – 5.b** Explain LVDT type accelerometer used for vibration measurement. **[05]****Q – 5.c** Explain variable capacitance transducer used for pressure measurement. **[06]****OR****Q – 5.c** Derive the equation for gauge factor of strain gauges. **[06]****Q – 6.a** Discuss the strain gauge type load cell in detail. **[06]****Q – 6.b** Explain strip chart recorder along with its advantages and disadvantages. **[07]****OR****Q – 6.b** Draw a schematic of McLeod Gauge used for vacuum pressure measurement. **[07]**
